10-7-26

Damped oscillator

(Underdamped, overdamped, critically damped)

$$F = \underbrace{-kx}_{F_{spring}} \underbrace{-b\frac{dx}{dt}}_{F_{damping}} = m\frac{d^2x}{dt^2}$$

$\frac{k}{m} = \omega_0^2 = (\text{natural freq})^2$, $\gamma \equiv \frac{b}{m}$ = damping factor

$$\ddot{x} + \gamma\dot{x} + \omega_0^2 x = 0$$

$\dot{x} = \frac{dx}{dt}$, $\ddot{x} = \frac{d^2x}{dt^2}$

$$x = Ae^{\lambda t}$$

$$\underbrace{(\lambda^2 + \gamma\lambda + \omega_0^2)}_{0}\underbrace{Ae^{\lambda t}}_{x_0} = 0 \quad \forall\, t \in (0,\infty) \qquad 0 \le t < \infty$$

Chr. eqn

$$\lambda^2 + \gamma\lambda + \omega_0^2 = 0$$

$$\left(\lambda + \frac{\gamma}{2}\right)^2 + \omega_0^2 - \left(\frac{\gamma}{2}\right)^2 = 0$$

(Case 1) $\omega_0 > \frac{\gamma}{2}$

$$\left(\lambda + \frac{\gamma}{2}\right)^2 = -\left[\omega_0^2 - \left(\frac{\gamma}{2}\right)^2\right]$$

$$\lambda = -\frac{\gamma}{2} \pm \underbrace{\sqrt{-1}}_{i}\underbrace{\left[\omega_0^2 - \left(\frac{\gamma}{2}\right)^2\right]^{1/2}}_{\omega}$$

$$x(t) = c_1 e^{-\frac{\gamma}{2}t + i\omega t} + c_2 e^{-\frac{\gamma t}{2}} e^{-i\omega t}$$

$$= e^{-\frac{\gamma t}{2}}\left[A'\cos\omega t + B'\sin\omega t\right)$$

$$x(t) = Ae^{-\gamma t}\cos(\omega t + \alpha)$$

$x(0) = x_0$ $\dot{x}(0) = v(0) = V_0$

$$x(0) = A\cos\alpha = x_0$$

$$x = Ae^{-\gamma t/2}\cos(\omega t + \alpha)$$

$$\dot{x} = -\frac{\gamma}{2}Ae^{-\frac{\gamma t}{2}}\cos(\omega t + \alpha) - \omega Ae^{-\frac{\gamma t}{2}}\sin(\omega t + \alpha)$$

$$\dot{x}(0) = -\frac{\gamma}{2}A\cos\alpha - \omega A\sin\alpha = V_0$$

Assume $V_0 = 0$

$$\frac{\gamma}{2}\cos\alpha + \omega\sin\alpha = 0$$

$$\tan\alpha = -\frac{\gamma}{2\omega}$$

$$\alpha = \tan^{-1}\left(-\frac{\gamma}{2\omega}\right) = -\tan^{-1}\left(\frac{\gamma}{2\omega}\right)$$

$x(0) = 0, \quad V(0) = V_0$

$A \cos\alpha = 0, \quad \alpha = \pm \pi/2$

$x(t) = A e^{-\frac{\gamma t}{2}} \cos(\omega t - \pi/2)$

$= A e^{-\frac{\gamma t}{2}} \sin \omega t$

$\dot{x} = -\frac{\gamma}{2} A e^{-\frac{\gamma t}{2}} \sin \omega t + A\omega e^{-\frac{\gamma t}{2}} \cos \omega t$

$\dot{x}(0) = A\omega = V_0$

$A = \frac{V_0}{\omega}$

$x = A e^{-\gamma t/2} \sin \omega t$

$\dot{x} = A e^{-\frac{\gamma t}{2}} \left[-\frac{\gamma}{2} \sin \omega t + \omega \cos \omega t \right]$

$\cos\varphi = \frac{-\gamma/2}{\sqrt{(\gamma/2)^2 + \omega^2}} = -\frac{\gamma}{2\omega_0}$

$\sin\varphi = \frac{\omega}{\sqrt{(\frac{\gamma}{2})^2 + \omega^2}} = \frac{\omega}{\omega_0}$

$(\frac{\gamma}{2})^2 + \omega^2 = (\frac{\gamma}{2})^2 + \omega_0^2 - (\frac{\gamma}{2})^2$

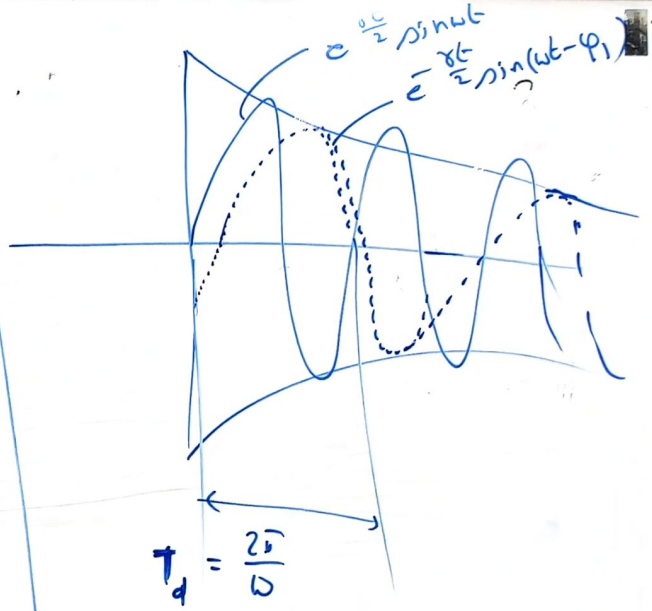
$\sqrt{(\frac{\gamma}{2})^2 + \omega^2} = \sqrt{\omega_0^2} = \omega_0$

$\dot{x} = A e^{-\frac{\gamma t}{2}} \left(\sin\omega t \cos\varphi + \cos\omega t \sin\varphi \right)$

$= A\omega_0 e^{-\frac{\gamma t}{2}} \sin(\omega t + \varphi) = A\omega_0 e^{-\frac{\gamma t}{2}} \sin(\omega t - \varphi_1)$

$\varphi = -\tan^{-1}\left(\frac{\gamma}{2\omega_0}\right) = -\varphi_1$

phase lag $\varphi_1 = \tan^{-1}\left(\frac{2\omega}{\gamma}\right)$



1st peak for $x(t)$ = $T_{p_1} = \frac{\pi/2}{\omega}$

" " " $V(t)$ = $\frac{\pi/2 + \varphi_1}{\omega}$

$+\langle k \rangle$

$m\omega_0^2 A^2 e^{-\gamma t}$

$\gamma \langle E \rangle$ = Powa

wa dissipated = $\gamma \langle E \rangle$

ity factor

$= \frac{2\pi \times E\text{- stored}}{E\text{- loss in one } T}$

$= 2\pi \frac{\langle E \rangle}{P \cdot T} = \frac{2\pi \langle E \rangle}{P/f}$

$= \frac{\omega \langle E \rangle}{P} = \boxed{\frac{\omega}{\gamma}}$

$$Q = \frac{\omega}{\gamma} \approx \frac{\omega_0}{\gamma}$$

$\omega_0 > \gamma/2$?

$$x = e^{-\frac{\gamma}{2}t}\left[c_1 e^{kt} + c_2 e^{-kt}\right]$$

$$k = \sqrt{\left(\frac{\gamma}{2}\right)^2 - \omega_0^2} \qquad \omega_0 < \frac{\gamma}{2}$$

Critically damped $\frac{\gamma}{2} = \omega_0$

$$x(t) = (A + Bt)\, e^{-\frac{\gamma t}{2}} \sin \omega t$$

Damped Oscillator

(Underdamped, overdamped, critically damped)

$$U = \frac{1}{2}kx^2 = \frac{1}{2}\underbrace{kA^2}_{m\omega_0^2} e^{-\gamma t}\sin^2\omega t$$

$$= \frac{1}{2}m\omega_0^2 A^2 e^{-\gamma t}\sin^2\omega t$$

$$\langle U\rangle \approx \frac{1}{4}m\omega_0^2 A^2 e^{-\gamma t} \quad \left(e^{-\gamma t} \approx \text{const over one } T\right)$$

$$v = \omega_0 A e^{-\frac{\gamma t}{2}}\sin(\omega t - \varphi_1)$$

$$\left\langle \frac{1}{2}mv^2\right\rangle \approx \frac{1}{2}m\omega_0^2 A^2 e^{-\gamma t}\langle \sin^2(\omega t-\varphi_1)\rangle$$

$$\approx \frac{1}{4}m\omega_0^2 A^2 e^{-\gamma t}$$

$$\langle E\rangle = \langle U\rangle + \langle K\rangle$$

$$\approx \frac{1}{2}m\omega_0^2 A^2 e^{-\gamma t}$$

$$\frac{d\langle E\rangle}{dt} = -\gamma\langle E\rangle = \text{Power}$$

P = Power dissipated $= \gamma\langle E\rangle$

Quality factor

$$= \frac{2\pi \times \text{E-stored}}{\text{E-loss in one } T}$$

$$= \frac{2\pi\langle E\rangle}{P\cdot T} = \frac{2\pi\langle E\rangle}{P/f}$$

$$= \frac{\omega\langle E\rangle}{P} = \boxed{\frac{\omega}{\gamma}}$$

$$\boxed{Q = \frac{\omega}{\gamma} \approx \frac{\omega_0}{\gamma}}$$

x =

Cr.

X